Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015118	Course Name:	Artificial Intelligence & Soft Computing Lab

Name of Student:	Rushikesh Subhash Khemnar
Roll No. :	81
Experiment No.:	6
Title of the Experiment:	Implement Supervised and Unsupervised Neural Network Learning Techniques
Date of Performance:	31/08/26
Date of Submission:	07/08/26

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : Signature :

Date :

Aim: To implement basic supervised and unsupervised neural network learning techniques for a given problem.

Objectives:

After completing this experiment, the student will be able to:

1. To understand the concepts of supervised and unsupervised learning in neural networks.
2. To implement a basic supervised learning technique for classification.
3. To implement a basic unsupervised learning technique for clustering.
4. To compare the working principles of supervised and unsupervised learning.
5. To analyze the performance of neural network learning techniques for different types of problems.

Tools Required:

- Hardware:
  - Desktop/Laptop
- Software:
  - Python 3.x
  - Visual Studio Code / PyCharm / Jupyter Notebook
  - NumPy Library
  - Scikit-learn Library
  - Windows/Linux Operating System

Dataset Details:

The experiment uses a simple dataset suitable for classification and clustering tasks. The dataset may consist of labelled data for supervised learning and unlabelled data for unsupervised learning to demonstrate the respective neural network learning techniques.

Theory:

Neural network learning techniques enable machines to learn patterns from data and make intelligent decisions. Based on the availability of labelled data, learning methods are broadly classified into Supervised Learning and Unsupervised Learning.

In Supervised Learning, the neural network is trained using labelled input-output pairs, allowing it to learn the relationship between the input features and the target output. It is commonly used for classification and regression problems. In Unsupervised Learning, the neural network is trained using unlabelled data, where it automatically discovers hidden structures or groups within the dataset. This approach is widely used for clustering, pattern recognition, and data exploration. Both learning techniques form the foundation of modern Artificial Intelligence and Machine Learning applications.

Procedure:

1. Select a suitable dataset for the experiment.
2. Preprocess the dataset by handling missing values and preparing the input data.
3. Divide the labelled dataset into training and testing sets for supervised learning.
4. Implement a supervised learning model and train it using the training data.
5. Implement an unsupervised learning technique using the unlabelled dataset.
6. Execute both learning models and observe the generated results.
7. Compare the outcomes of supervised and unsupervised learning and record the observations.

Description of the Experiment:

This experiment demonstrates the implementation of supervised and unsupervised neural network learning techniques using a suitable dataset. A supervised learning model is trained using labelled data to perform prediction or classification, while an unsupervised learning model is applied to identify hidden patterns or clusters in unlabelled data.

The experiment provides an understanding of the differences between the two learning approaches and their applications in solving real-world Artificial Intelligence problems.

Detailed Description:

Supervised learning is performed by training a neural network using labelled data, where the expected output is known for each input sample. The model learns from these examples and predicts the output for new data based on the learned patterns. In contrast, unsupervised learning is carried out using unlabelled data, where the model identifies similarities among data points and groups them into meaningful clusters without prior knowledge of the output.

This experiment involves implementing one supervised and one unsupervised learning technique on a suitable dataset and comparing their behaviour, learning process and outputs. The observations obtained from both implementations help in understanding the suitability of each learning approach for different Artificial Intelligence applications.

Code:-

```
import numpy as np import
matplotlib.pyplot as plt

from sklearn.datasets import load_iris from
sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import
MLPClassifier from sklearn.cluster import
KMeans from sklearn.metrics import (
accuracy_score,     classification_report,
confusion_matrix,
    silhouette_score
)

iris = load_iris()

X = iris.data
y = iris.target

print("Dataset loaded successfully.")
print("Number of samples:", X.shape[0])
print("Number of features:", X.shape[1])
print("Classes:", iris.target_names)

print("\nMissing values:", np.isnan(X).sum())

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print("Data preprocessing completed.")

print("\n========== SUPERVISED LEARNING ==========")

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled,
    y,
    test_size=0.20,
random_state=42,
stratify=y
)

model = MLPClassifier(
hidden_layer_sizes=(10,),
activation='relu',
solver='adam',
max_iter=1000,
    random_state=42
```

```
)

model.fit(X_train, y_train)

print("Supervised neural network training completed.")

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("\nSupervised Learning Accuracy:",
      round(accuracy * 100, 2), "%")

print("\nClassification Report:")
print(classification_report(
   y_test,     y_pred,
target_names=iris.target_names
))

cm = confusion_matrix(y_test, y_pred)

print("Confusion Matrix:")
print(cm)

print("\n========== UNSUPERVISED LEARNING ==========")

# Create K-Means model
kmeans = KMeans(
   n_clusters=3,
random_state=42,
   n_init=10
)

clusters = kmeans.fit_predict(X_scaled)

print("Unsupervised clustering completed.")

print("\nCluster assignments:")
print(clusters)

sil_score = silhouette_score(X_scaled, clusters)
```

```
print("\nSilhouette Score:",
      round(sil_score, 4))

print("\n========== CLUSTER COMPARISON ==========")

print("\nActual class labels:")
print(y)

print("\nPredicted cluster labels:")
print(clusters)


plt.figure(figsize=(8, 6))

plt.scatter(
X_scaled[:, 0],
X_scaled[:, 1],
c=clusters,
cmap='viridis',
    s=50
)

plt.scatter(
kmeans.cluster_centers_[:, 0],
kmeans.cluster_centers_[:, 1],
    c='red',
marker='X',
s=200,
    label='Cluster Centers'
)

plt.xlabel("Sepal Length (Standardized)")
plt.ylabel("Sepal Width (Standardized)") plt.title("K-
Means Clustering of Iris Dataset") plt.legend()
plt.grid(True)

plt.show()

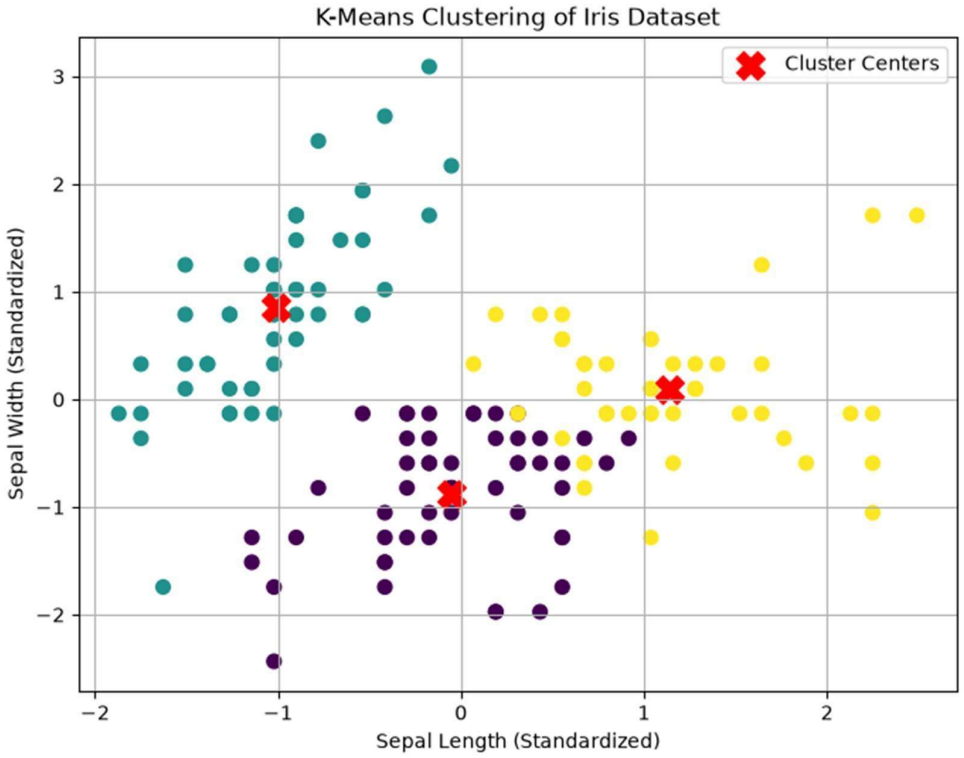

print("\n========== FINAL COMPARISON ==========")
```

```
print("Supervised Learning:") print("- Technique:
Multi-Layer Perceptron (MLP)")
print("- Task: Classification")
print("- Uses labelled data: Yes")
print("- Accuracy:", round(accuracy * 100, 2), "%")

print("\nUnsupervised Learning:")
print("- Technique: K-Means
Clustering")
print("- Task: Clustering")
print("- Uses labelled data: No")
print("- Number of clusters: 3")
print("- Silhouette Score:", round(sil_score, 4))

print("\nExperiment completed successfully.")
```

Output:-



```
4\bundled\libs\debugpy\launcher 60540 -- C:\Users\student\Downloads\mw.py "
Dataset loaded successfully.
Number of samples: 150
Number of features: 4
Classes: ['setosa' 'versicolor' 'virginica']

Missing values: 0
Data preprocessing completed.

========== SUPERVISED LEARNING ==========
Supervised neural network training completed.

Supervised Learning Accuracy: 96.67 %

Classification Report:
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       1.00      0.90      0.95        10
   virginica       0.91      1.00      0.95        10

    accuracy                           0.97        30
   macro avg       0.97      0.97      0.97        30
weighted avg       0.97      0.97      0.97        30

Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  0 10]]
```

```
========== UNSUPERVISED LEARNING ==========
Unsupervised clustering completed.

Cluster assignments:
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 0 0 0 2 0 0 0 0 0 0 0 2 0 0 0 0 2 0 0 0
 0 2 2 2 0 0 0 0 0 0 0 2 2 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 2 2 2 2 0 2 2 2 2
 2 2 0 0 2 2 2 2 0 2 0 2 0 2 2 0 2 2 2 2 2 2 2 0 0 2 2 2 0 2 2 2 0 2 2 2 0 2
 2 0]

Silhouette Score: 0.4599

========== CLUSTER COMPARISON ==========

Actual class labels:
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2
 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
 2 2]

Predicted cluster labels:
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 0 0 0 2 0 0 0 0 0 0 0 2 0 0 0 0 2 0 0 0
 0 2 2 2 0 0 0 0 0 0 0 2 2 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 2 2 2 2 0 2 2 2 2
 2 2 0 0 2 2 2 2 0 2 0 2 0 2 2 0 2 2 2 2 2 2 2 0 0 2 2 2 0 2 2 2 0 2 2 2 0 2
 2 0]
```

```
========== FINAL COMPARISON ==========
Supervised Learning:
- Technique: Multi-Layer Perceptron (MLP)
- Task: Classification
- Uses labelled data: Yes
- Accuracy: 96.67 %

Unsupervised Learning:
- Technique: K-Means Clustering
- Task: Clustering
- Uses labelled data: No
- Number of clusters: 3
- Silhouette Score: 0.4599

Experiment completed successfully.
```

Conclusion:

Basic supervised and unsupervised neural network learning techniques were successfully implemented and analyzed for a given problem. The experiment demonstrated the differences in their learning mechanisms and their applicability to classification and clustering tasks.

Questions to be Answered by Students:

1. Which learning technique was implemented using labelled data?

   Supervised learning was implemented using labelled data. A Multi-Layer Perceptron (MLP) neural network was used to classify the Iris flowers into three classes: Setosa, Versicolor, and Virginica.

2. Which learning technique was used to identify patterns in unlabelled data?

   Unsupervised learning was used to identify patterns in unlabelled data. K-Means clustering was used to group the Iris data into three clusters based on similarities between the features.

3. State two differences observed between supervised and unsupervised learning techniques.

   Supervised learning uses labelled data with known output classes, whereas unsupervised learning uses unlabelled data without known output classes.

Supervised learning is mainly used for prediction and classification, whereas unsupervised learning is used to discover hidden patterns and groups in the data.

4. What observations were obtained from the supervised and unsupervised learning models?

The MLP supervised model successfully learned the relationship between the input features and flower classes and achieved high classification accuracy.
The K-Means unsupervised model successfully grouped the Iris samples into three clusters based on feature similarities, without using the actual class labels during training.
Thus, supervised learning is more suitable when the target output is known, while unsupervised learning is useful for discovering unknown patterns or groups.